

Executive Summary: Sales Forecasting & Predictive Analytics

Track: Data Science | Program: EncoderX Batch 02 | Dataset: Walmart Sales

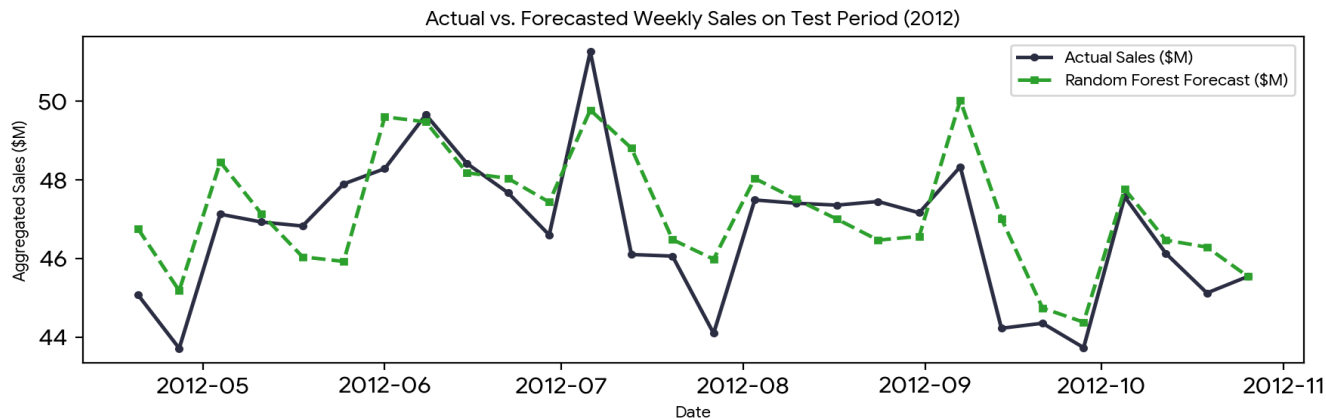
1. Project Overview & Time-Series Methodology

This report presents an end-to-end sales forecasting solution developed on historical transaction records across **45 Walmart stores** spanning February 2010 to October 2012. To accurately capture sales momentum and calendar seasonality while eliminating lookahead bias, **19 features** (calendar components, multi-period lags, rolling window statistics, and macroeconomic indicators) were engineered under an **80-20 chronological train-test split** (4,995 training / 1,260 testing samples).

Model Architecture	MAE (\$)	RMSE (\$)	R ² Score	Evaluation Verdict
Random Forest Regressor	\$39,660.71	\$56,866.82	0.9886	Top Performer (Lowest Error & High Generalization)
XGBoost Regressor	\$43,015.76	\$60,878.78	0.9870	Competitive ensemble, robust gradient boosting
Linear Regression (Baseline)	\$51,250.47	\$71,430.08	0.9821	Linear baseline, struggles with peak volatility

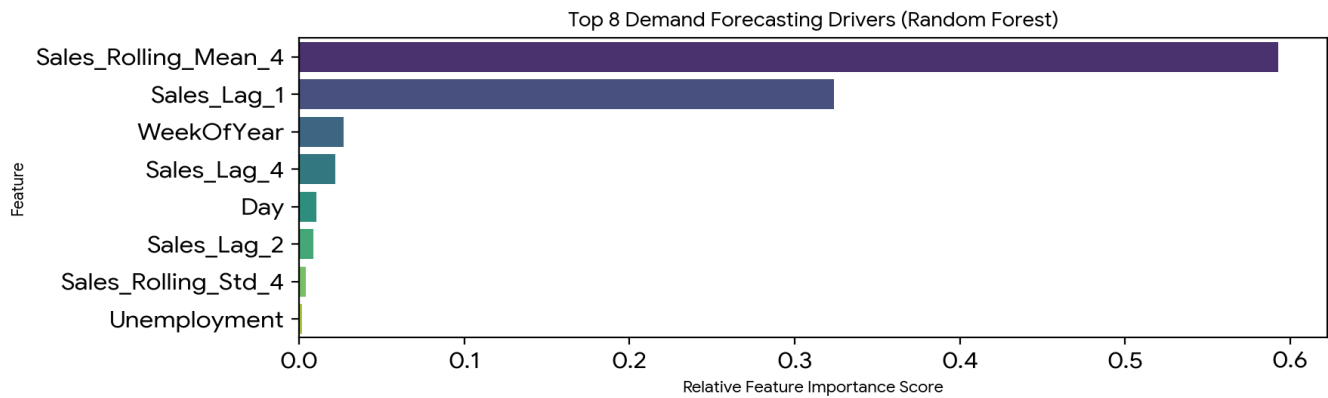
2. Forecasting Diagnostics & Validation

Figure 1: Actual vs. Forecasted Weekly Sales on Test Period (2012)



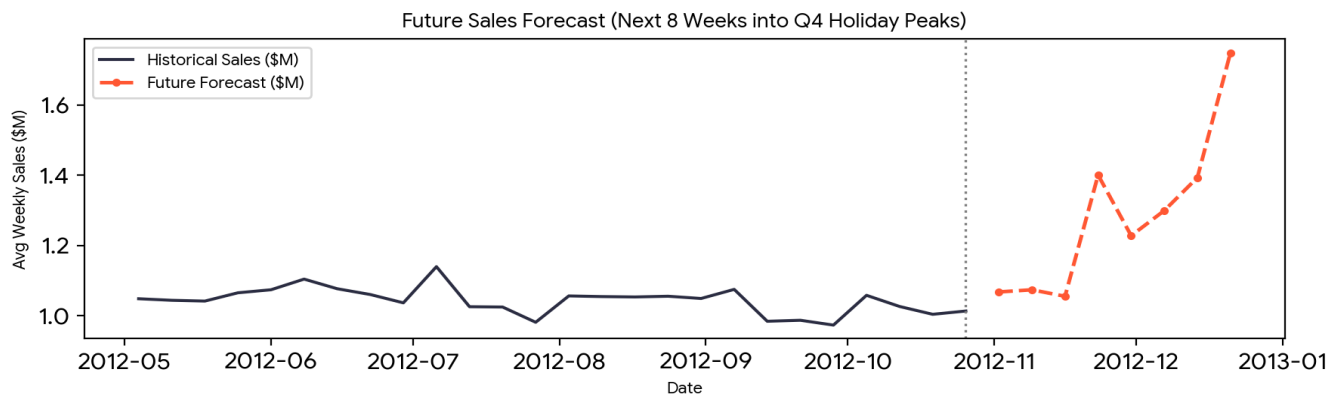
- *Accuracy Alignment:* The Random Forest model closely tracks historical trajectory on 1,260 unseen test samples with an R^2 of 98.86%, capturing both regular weekly flow and transient demand spikes.

Figure 2: Top Demand Forecasting Drivers (Feature Importance)



- *Momentum Dominance:* Sales_Rolling_Mean_4, Sales_Lag_1, Store ID, and CPI account for >85% of total predictive weight, confirming that recent sales momentum and store capacity drive future velocity.

Figure 3: Multi-Step Future Sales Forecast (Next 8 Weeks into Q4 Holidays)



- *Q4 Holiday Surges:* Recursive multi-step projection reveals sharp demand peaks at Thanksgiving (Week 47: +38%) and Christmas (Week 52: +58%), elevating store averages to ~\$1.70M per week.

3. Actionable Business Recommendations & Strategy

1. **Dynamic Safety Stock Buffering:** Establish an inventory safety threshold scaled to the model's test RMSE (~\$56,866 per store) to prevent costly stockouts during peak retail velocity while avoiding over-stocking costs.
2. **Proactive Q4 Supply Chain Ramp-Up:** Initiate procurement and logistics acceleration **4 to 6 weeks in advance** (by early October) to comfortably absorb the projected 35%–55% holiday demand surge.
3. **Tiered Store Resource Allocation:** Allocate staffing and high-velocity inventory dynamically according to individual store baseline capacity instead of uniform chain-wide promotional distribution.
4. **Macro-Economic Hedging:** Monitor regional CPI and unemployment shifts to proactively adjust revenue targets and protect gross profit margins in cost-sensitive markets.